

COMP.SE.140 final project

Instructions for the teaching assistant

1. Clone the repository

```
$ git clone -b project https://compse140.devops-gitlab.rd.tuni.fi/pgmaha/final-project.git
```

```
$ cd <created folder>
```

```
$ docker-compose build --no-cache
```

```
$ docker-compose up -d
```

Optionally: https://github.com/Anfood/Docker_compose_exercise.git

Implemented optional features

- None

Instructions for examiner to test the system.

- On the browser go to <http://localhost:8198/>
 - o Login using login.txt
 - o Press REQUEST a bunch of times
- On the browser for example <http://localhost:8197/state> gives the current state
 - o You can see the PAUSED state when you click on the html button 10 times in quick succession and the refresh <http://localhost:8197/state>
 - o You can use curl "localhost:8197/run-log" -X GET for state change history

Data about the platform you used in development

- Windows 10
- Docker version 25.0.3, build 4debf41
- Docker Compose version v2.24.6-desktop.1
- Desktop PC
 - o CPU: x64-based AMD Ryzen 9 5900X
 - o Geforce RTX 3080

Description of the CI/CD pipeline

- 3 stages
 - o Build
 - Builds the Docker images
 - o Test
 - Runs own test scripts for the API endpoints
 - o Deploy
 - Deploys the services using Docker Compose
- Gitlab Runner is registered and runs locally

Briefly document all steps:

- Version management; use of branches etc
 - The same code exists both in the aforementioned GitLab and Github, in their “project” branch
- Building tools
 - Visual studio code
- Testing; tools and test cases
 - Testing the API endpoints with simple Linux shell commands and calls, calling the docker API to access the containers from the Host on Gitlab
- Packing
 - Docker compose was used to manage services
- Deployment
 - Services are deployed using Docker Compose
- Operating; monitoring
 - No monitoring implemented

Example runs of the pipeline

Filter pipelines			
Status	Pipeline	Created by	Stages
✓ Passed ⌚ 00:02:12 📅 18 minutes ago	Removing the shutdown confirmation test as I ... #4022 project 74c6410d latest		✓-✓-✓
✗ Failed ⌚ 00:01:12 📅 25 minutes ago	Fixing run-log #4014 project 6cb6dd80		✓-✗-»
✗ Failed ⌚ 00:01:13 📅 33 minutes ago	Made service1 change the state via API #4010 project 5ba24a7f		✓-✗-»
✗ Failed ⌚ 00:01:11 📅 55 minutes ago	Added more state handling #4001 project 9ef06f58		✓-✗-»
✗ Failed ⌚ 00:01:19 📅 1 hour ago	Fixing test syntax and adding changing state f... #3958 project a6ca18d2		✓-✗-»
✗ Failed ⌚ 00:01:11 📅 2 hours ago	Fixed the test syntax #3949 project beb8160c		✓-✗-»
✗ Failed ⌚ 00:01:10 📅 2 hours ago	Making rest of the state tests #3942 project 13b2af2d		✓-✗-»
✓ Passed ⌚ 00:02:09 📅 4 hours ago	making the first state fetch test to fail #3900 project 5c6e7d62		✓-✓-✓
✓ Passed ⌚ 00:02:13 📅 1 day ago	Implementing API GET method port 8197 #3684 project 54646af1		✓-✓-✓
✓ Passed ⌚ 00:02:01 📅 1 day ago	Fixed variable value setting syntax #3530 project 57e8e4a2		✓-✓-✓
✗ Failed ⌚ 00:00:46	Testing yet another url to access the Docker se... #3529 project 037ffc58		✓-✗-»

Include some kind of log of both failing test and passing.

Failing tests example:

```
307 Container service3_nginx Waiting
308 Container service4_rest-api Waiting
309 Container service2_python-backend Healthy
310 Container service4_rest-api Healthy
311 Container final-project-service1-2 Healthy
312 Container final-project-service1-1 Healthy
313 Container final-project-service1-3 Healthy
314 Container service3_nginx Healthy
315 $ chmod +x ./tests/test.sh
316 $ chmod +x ./tests/test_api.sh
317 $ ./tests/test.sh
318 Running Hello test
319 Hello test passed
320 $ ./tests/test_api.sh
321 ✓ Test passed: Received 200 OK for /GET state
322 ./tests/test_api.sh: line 20: [INIT: not found
323 ✗ Test failed: State is not INIT
324 State: INIT
325 ✗ Test failed: State is not RUNNING
326 State: 404
327 ./tests/test_api.sh: line 35: [404: not found
328 ✗ Test failed: State is not PAUSED
329 State: 404
330 ./tests/test_api.sh: line 50: [404: not found
331 ✗ Test failed: Containers are not shut down
332 State: 404
333 ./tests/test_api.sh: line 65: [404: not found
334 ✗ Test failed: Docker containers are not shut down
335 State: 37228c625a37
336 7b40d4ffb8a9
337 f141c8c7e259
338 67f6bee7b7b6
339 0394e60f1636
340 7ee980326177
341 ./tests/test_api.sh: line 77: [37228c625a37
342 7b40d4ffb8a9
343 f141c8c7e259
344 67f6bee7b7b6
345 0394e60f1636
346 7ee980326177: not found
✓ 347 Cleaning up project directory and file based variables
348 ERROR: Job failed: exit code 1
```

Passing tests example:

```
326 Container final-project-service1-3 Healthy
327 $ chmod +x ./tests/test.sh
328 $ chmod +x ./tests/test_api.sh
329 $ ./tests/test.sh
330 Running Hello test
331 Hello test passed
332 $ ./tests/test_api.sh
333 ✓ Test passed: Received 200 OK for /GET state
334 ✓ Test passed: State is INIT
335 ✓ Test passed: Received 200 OK for /PUT 'RUNNING' state
336 ✓ Test passed: Received 200 OK for /PUT 'PAUSED' state
337 ✓ Test passed: Received 200 OK for /GET request
338 ✓ Test passed: Received 200 OK for /PUT 'SHUTDOWN' state
339 $ docker compose down
340 Container service3_nginx Stopping
341 Container service3_nginx Stopped
342 Container service3_nginx Removing
343 Container service3_nginx Removed
344 Container service4_rest-api Stopping
345 Container service4_rest-api Stopped
346 Container service4_rest-api Removing
```

Reflections

This was incredibly difficult project. There was so much to learn, and it's a shame that although I spent a lot of time on this, I didn't manage to finish it. But I'm happy about what I was able to achieve.

Main learnings and worst difficulties

I learned how to make multiple services work together to form a complete application using Docker. I was very happy to achieve a multi container application like this that does cool simple things. The CI/CD pipeline was more comprehensive than in my earlier testing course.

It was incredibly hard to break down the project requirements into sub-problems and then to go through hundreds of websites to look for different kinds of solutions. I sure do have programming knowledge, but I needed to learn so much more and integrate so many solutions to each other which was difficult.

I left out implementing the shutdown of Docker containers from within a container because that proved too difficult to me. I also didn't implement the extra features as I ran out of time.

Amount effort (hours) used

40 hours